

HIGH LEVEL DESIGN (HLD)

1. Introduction

The RAG-Based Customer Support Assistant is an AI-powered system designed to automate customer service interactions using Retrieval-Augmented Generation (RAG). The system leverages a predefined knowledge base in the form of a PDF document to answer customer queries related to refund policies, shipping details, cancellations, and frequently asked questions.

Traditional customer support systems rely heavily on human agents, which leads to increased operational costs and slower response times. This project aims to improve efficiency by introducing an intelligent assistant capable of understanding user queries and generating accurate responses based on stored knowledge.

2. Objective

The primary objectives of this system are:

- ✚ To automate customer support responses using AI
- ✚ To provide accurate and context-aware answers
- ✚ To reduce dependency on manual support agents
- ✚ To implement human-in-the-loop escalation for sensitive queries
- ✚ To ensure scalability and modularity in design

3. System Overview

The system follows a Retrieval-Augmented Generation architecture. Instead of generating responses purely from a language model, it first retrieves relevant information from a structured knowledge base.

The knowledge base is stored as a PDF document containing customer service policies. This document is processed, split into smaller chunks, and converted into vector embeddings. These embeddings are stored in a vector database.

When a user submits a query, the system retrieves the most relevant chunks and passes them to the language model to generate a response.

4. Architecture Description

The architecture consists of the following flow:

-  User inputs a query
-  Query is processed and converted into embedding
-  System searches vector database for similar content
-  Relevant chunks are retrieved
-  Retrieved data is passed to LLM
-  LLM generates response
-  If query is sensitive, it is escalated

5. System Components

User Interface

Allows users to input queries and view responses.

Knowledge Base

A PDF document containing structured customer support information.

Embedding Model

The model sentence-transformers/all-MiniLM-L6-v2 is used to convert text into numerical vectors.

Vector Database

ChromaDB stores embeddings and enables semantic search.

Retriever

Fetches the most relevant chunks based on similarity.

Language Model

The llama-3.1-8b-instant model via Groq API generates responses.

Lang Graph Workflow

Controls execution flow and handles conditional routing.

6. Data Flow

The data flows through the system as follows:

User Query → Query Embedding → Vector Search → Retrieve Chunks → LLM Processing → Response Generation → Escalation (if needed)

7. Assumptions and Limitations

System works only with predefined PDF knowledge

Accuracy depends on quality of embeddings

No real-time data updates

Limited handling of ambiguous queries